

BANARAS HINDU UNIVERSITY

B.Voc. Semester I Examination, 2022-23

Paper: BVCA-103 — Applied Mathematics

Time: Three Hours

Maximum Marks: 70

Instructions: Answer all Sections. Candidates are required to solve their answers with accuracy.

SECTION — A: Long Answer Questions

[15 × 2 = 30]

Instructions: Long answer type questions each have to be solved stepwise.

Question 1. Prove that:

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$$

OR**Question 2.** Apply Gauss elimination method to solve the equations:

$$x + 4y - z = -5$$

$$x + y - 6z = -12$$

$$3x - y - z = 4$$

Question 3. Find the eigenvalues and verify the Cayley-Hamilton theorem for the given matrix:

$$A = \begin{pmatrix} 5 & 4 \\ 1 & 2 \end{pmatrix}$$

OR**Question 4.** Define with examples:

(a) The following:

(i) Set

(ii) Singleton Set

(iii) Finite Set

(iv) Infinite Set

(v) Null Set

(vi) Subset

- (b) If $n(A) = 50$, $n(B) = 60$, and $n(C) = 25$, then find $n(A \cup B \cup C)$ if A, B, C are disjoint sets.

SECTION — B

Note: Attempt any six out of eight. Short answer type questions to be solved in required steps: (5 × 6 = 30)

Question 1. Calculate the number of subsets for the given set $A = \{0, 1, 2\}$ and write all the possible subsets of the given set.

Question 2. Find the probability of getting Head and Tail both on one toss of a single coin.

Question 3. Write the characteristic equation of the given matrix:

$$A = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix}$$

Question 4. Write the statement of Cayley-Hamilton theorem and verify this theorem for an Identity matrix of order 3×3 .

Question 5. Define Derivation Tree.

Question 6. Apply Gauss-Seidel iteration method to solve the equations:

$$\begin{aligned} 20x + y - 2z &= 17 \\ 3x + 20y - z &= -18 \\ 2x - 3y + 20z &= 25 \end{aligned}$$

Perform this operation up to two iterations.

Question 7. Define set of set and write the power set of the given set $A = \{1, 2, 3\}$.

Question 8. Define the sample space and write the sample space of the outcome of a throw of a dice, and find the probability that we get 6 in a single throw of dice.

SECTION — C

Note: Attempt all questions. Very short answer type questions or one-word answer type questions: (10 × 1 = 10)

1. Can we calculate the determinant value of a row matrix?
2. Verify if the Identity matrix of order 2×2 is a non-singular matrix.
3. Why is the system of Homogeneous equations called consistent?
4. Can we refer to a null set as an empty set?

5. Calculate the value of the given determinant:

$$A = \begin{vmatrix} 1 & 0 & 1 \\ 1 & 0 & 2 \\ 1 & 1 & 2 \end{vmatrix}$$

6. Write what type of set is $A = \{\}$.

7. Write the definition of 'Mutually Exclusive Event'.

8. Calculate the number of terms in the expansion of a 3×3 determinant.

9. What is an Upper Triangular Matrix?

10. Define Function.